

Supermarket Sales Dashboard Analysis

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**Domain: Data Analytics & Business Intelligence (Excel + SQL +
Power BI/ Tableau)**

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Task-2: Advanced Data Analysis, SQL Joins & Business KPI Dashboard.

- **Project Overview:**

This project focuses on analyzing supermarket sales data using SQL, Excel, and Power BI. The objective of the project was to identify business performance trends, customer behavior, profit analysis, and regional sales insights through data analytics and visualization techniques. The project included database creation, SQL JOIN operations, KPI calculations, dashboard development, and business performance analysis.

- **Tools Used:**

Tools	Purpose
Excel	Data Cleaning, Analysis & Pivot Tables
DB Browser (SQLite)	Data Extraction, SQL Queries
Power BI	Dashboard Visualization

- **Business Performance Analysis:**

1. **Which region generates the highest revenue?**

The **West Region** generated the highest revenue among all regions. This indicates strong customer demand and better sales performance in that area. The region contributed significantly to overall business growth and profitability.

2. **Which category has the highest profit margin?**

The **Technology Category** achieved the highest profit margin. Technology products generated better profit compared to Furniture and Office Supplies. This category plays an important role in increasing overall business profit.

3. **Are discounts reducing profitability?**

Yes, higher discounts are reducing profitability in several transactions. Orders with large discount percentages showed lower profit margins. This indicates that discount strategies should be managed carefully to avoid profit loss.

4. **Which customer segment is most valuable?**

The **Consumer Segment** is the most valuable customer segment. It contributed the highest number of orders and generated strong sales revenue. The segment plays a major role in overall business performance.

5. **Is there a seasonal sales trend?**

Yes, the business shows a clear seasonal sales trend. Sales increased during the final months of the year, especially in **November and December**. This suggests higher customer purchasing activity during festive and holiday seasons.

- **Excel Analysis:**

- **Pivot Tables:**

- **Sales By Region:**

Sales By Region	
Row Labels	Sum of sales
Central	345992.9567
East	397446.0149
South	286087.2943
West	478948.2983
Grand Total	1508474.564

- **Profit By Category & Region:**

Profit By Category & Region	
Row Labels	Sum of profit
Central	69198.59134
Furniture	22315.40814
Office Supplies	21131.6874
Technology	25751.4958
East	79489.20298
Furniture	23865.95338
Office Supplies	24489.526
Technology	31133.7236
South	57217.45886
Furniture	15927.79226
Office Supplies	18267.5546
Technology	23022.112
West	95789.65966
Furniture	35756.47646
Office Supplies	30105.8112
Technology	29927.372
Grand Total	301694.9128

- **Monthly Sales Trend:**

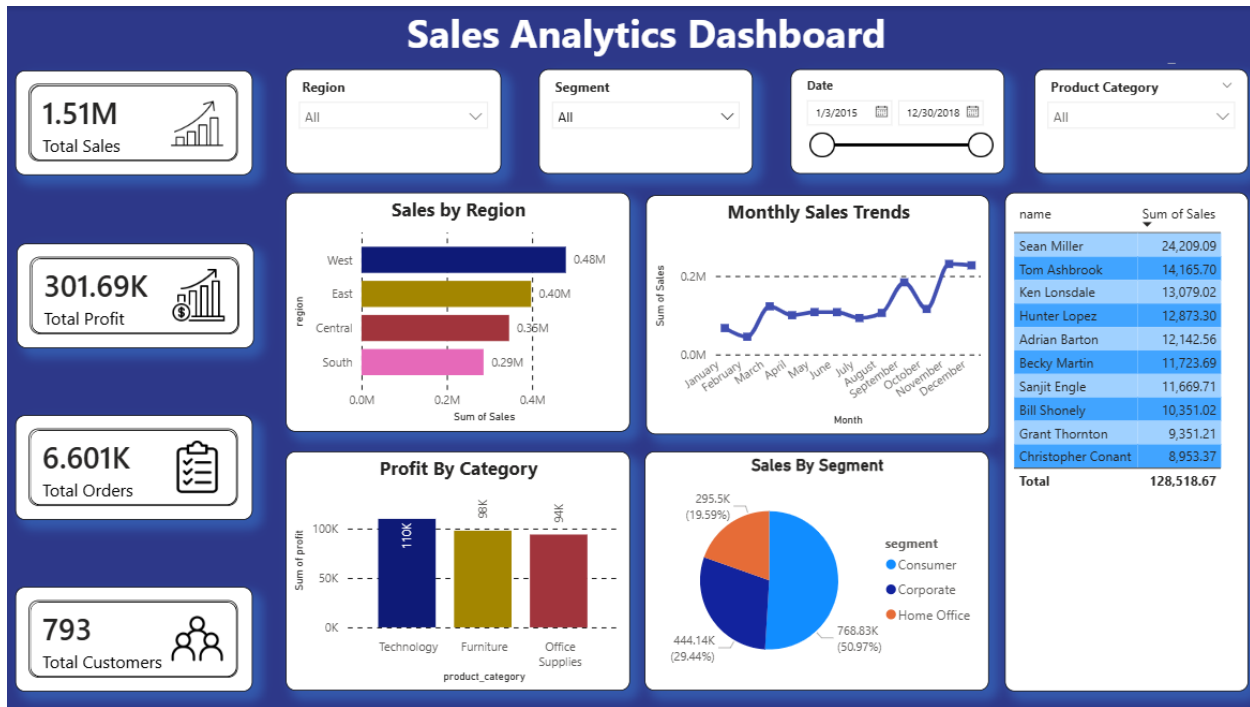
Monthly Sales Trend	
Row Labels	Sum of sales
January	110978.366
February	71900.7864
March	135537.2138
April	103268.6731
May	116974.6719
June	101263.6485
July	103965.7915
August	141710.0421
September	147032.2641
October	119715.022
November	180986.3118
December	175141.773
Grand Total	1508474.564

- **TOP-10 Customers:**

Top-10 Customers	
Row Labels	Sum of sales
Adrian Barton	12142.555
Becky Martin	11723.69
Bill Shonely	10351.023
Christopher Conant	8953.368
Grant Thornton	9351.212
Hunter Lopez	12873.298
Ken Lonsdale	13079.015
Sanjit Engle	11669.714
Sean Miller	24209.094
Tom Ashbrook	14165.7
Grand Total	128518.669

- **Power BI Dashboards:**

Screenshot:



[Fig. 1: Sales Analytics Dashboard]



[Fig. 2: Filtered Sales Analytics Dashboard]

- **KPI Overview:**

The dashboard provides important business KPIs including:

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin

These KPIs help measure overall business performance and support better decision-making.

- **Regional Performance:**

The regional analysis showed that the West Region generated the highest sales revenue. The East and Central regions also contributed strong sales performance, while some regions showed comparatively lower profitability.

Regional comparison helped identify high-performing business locations and growth opportunities.

- **Category Performance:**

Among all product categories, **Technology** generated the highest profit margin. Furniture contributed strong sales revenue, while Office Supplies maintained stable performance.

Category analysis helped understand which products generate better profitability for the business.

- **Customer Analysis:**

The Consumer segment contributed the highest number of orders and overall revenue. Top customers generated significant sales value, showing the importance of customer retention and loyalty.

Customer analysis also helped identify valuable customer segments for future business strategies.

- **Seasonal sales Trend:**

Sales performance increased during the final months of the year, especially in November and December. This indicates seasonal demand and higher purchasing activity during festive periods.

The monthly sales trend analysis helped identify peak business periods.

- **Conclusion:**

This project successfully demonstrated the use of SQL, Excel, and Power BI for business intelligence and sales analytics. The dashboard and analysis provided meaningful insights into sales performance, profitability, customer behavior, and business trends.

The project enhanced practical skills in:

- SQL JOIN operations
- KPI calculations
- Data visualization
- Dashboard design
- Business analytics